Software Requirements Specification

for

Project CommonGround

Version 1.0

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1. Introduction

1.1 Purpose

The purpose of this document is to specify the software requirements for Project CommonGround, a website application that allows users to buy and sell used items in a more secure manner than what already exists. The application services users in Harris County.

This website application is for people to find thrifted items online, similar to apps like eBay or Marketplace, with an emphasis on local pickup. We can use the 2 user's addresses to suggest 'safe' pickup locations to meet at, like grocery store parking lots, public parks, etc.

1.2 Intended Audience and Reading Suggestions

The intended audience of this document are developers, any potential testers, and Dr. David Burris.

1.3 Product Scope

Project CommonGround will exist to serve as a safer alternative to Craigslist and FB Marketplace which can be scam-heavy. The main goal is to simplify buying used items directly instead of using a third-party app. All we will do is allow people to list their clothing/items, process payment, and let them decide where to meet up. We will process payments through the app and find safe local meetup spots. We will not handle shipping details for users because in person meetups are encouraged between users.

1.4 References

This app was inspired by other self-listing selling platforms such as

- Mercari: www.mercari.com/, accessed 2-2026
- FB Market: https://www.facebook.com-/marketplace/, accessed 2-2026
- eBay: www.ebay.com/, accessed 2-2026
- Craigslist: craigslist.org, accessed 2-2026
- Depop: www.depop.com/, accessed 2-2026

2. Overall Description

2.1 Product Perspective

This is a new self contained product that is not a component of a larger system. It is not a part of a product family, and it could be a replacement for other used item sale platforms. It aims to promote buying used items within a local community. This is good for people who are hesitant to use other platforms because of the potential for high scam rates.

2.2 Product Functions

- Account creation
    - New user or log-in
    - Authentication (email / phone number)
    - Address
- Product listing
    - Product Creation
- Product Buying
    - Payment Processing
- Map Meetup
    - A single map that allows users to select/request meetup areas.
    - Save favorite locations

2.3 User Classes and Characteristics

Primary users:

- Buyer / seller: The system shall allow all features related to making a purchase and making a sale, including adding an item to cart, leaving ratings, and coordinating pickups once an account is made within the system.

Secondary users:

- Admin accounts: The system shall give administrative accounts all permissions in order to do proper maintenance such as removing fraudulent listings, suspending users, viewing reports, and performing updates. Admins are not permitted to buy or sell using their admin profile. They must use a personal account.
- Guest Account: The system shall only allow guests accounts to browse listings.

2.4 Operating Environment

- The system shall function on a web-based interface as a website.
- The system shall support all browsers such as Chrome, Firefox, Safari, etc.
- The system shall be compatible on PCs and phones regardless of what OS (Apple, Android, Windows, etc).

2.5 Design and Implementation Constraints

Security:

- Refunds
- Lost Data: Recovery and Backups for database
- Home addresses not publicly displayed
- Hashing passwords

Hardware and Software Limitations:

- There is a limitation on outlining exact performance details as this product is in the process of being developed.
- Memory relies on a singular device (no large server availability).

Buyer/Seller Interactions (Race Conditions):

- When a seller is attempting to update/delete a listing while a buyer tries to purchase at the same time.
- Two buyers trying to purchase the same item at the same time.
- Developers will be responsible for maintenance
    - MTTR: 1 day
    - MTBF: 4 Months (Increased traffic between seasons)

2.6 User Documentation

When the user creates an account, the user shall be redirected to an onboarding page to show the user how to use our platform.

2.7 Assumptions and Dependencies

- Users should be able to and willing to travel to a middle point between them, and can have a radius based on a saved address.
- We are not legally responsible for anything that goes wrong (user confirmation on both sides are both included)(user identity accuracy).
- The users should be able to access the Internet on their mobile devices and from their desktops.
- The system shall make use of map API to create a meetup feature.
- Quality Inspection: When meeting in person, a user is expected to bear the burden of checking the product for quality and making sure everything works properly before confirming the sale.

3. External Interface Requirements

3.1 User Interfaces

- Customer Web Portal
    - CAN view as guest but CANNOT add to cart/purchase
    - Home page displaying featured and recent listings
    - Search bar with filtering options (category, price range, distance)
    - Account registration and login screens
    - Product listing,creation, and editing pages - for signed in users only
    - Checkout/payment screen - for signed in users only
    - Interactive map for meetup selection - for signed in users only
    - Ratings and review interface - for signed in users only
    - Message seller - for signed in users only
- Local sale homepage
    - Consistent navigation bar across all pages
    - Clear error message display in red text beneath invalid fields
    - Required fields marked with an asterisk (*)
    - Confirmation messages displayed after successful transactions
    - Responsive layout compatible with standard desktop screen resolutions
    - Cart feature for users to add items they want to purchase
    -
- Administrator Dashboard -
    - User account management tools
    - Listing moderation controls
    - Fraud report viewing page

    - Transaction log access
    - User suspension and reinstatement functionality

3.2 Software Interfaces

- User Locations to Map API (pull addresses to connect to the map)
- User-to-User messaging interface
- Data storage interface (User accounts, listing, rating reports, etc)
- Back-up recovery interface
- Buyer and seller confirmation

3.3 Communications Interfaces

- Communication Protocols
    - Mobile and web browser app: HTTPS (TLS encryption required), DNS, and TCP (IP networking).
    - Used to communicate purchases and maintenance to users.

4. System Features

4.1 Payment Processing

4.1.1 Description and Priority

If a user intends to purchase an item, the app holds payment until after both sides meet in-person and confirm through the app that they have exchanged the item, then pushes the payment through. High priority.

4.1.2 Stimulus/Response Sequences

- Payment authorized → System reserves listing → Status updated to "reserved"
- Buyer selects "Purchase" → System verifies listing availability → Payment authorization requested
- Buyer confirms item reviewed → System records confirmation
- Seller confirms item delivered → System records confirmation
- Both confirmations present → System marks transaction "Completed" → Payment released to seller
- Payment authorization fails → Error message → Aboard transaction

4.1.3 Functional Requirements

REQ-1: System requires payment authorization before reserving a listing.
REQ-2: System will holds funds until transaction is complete and authorized.

REQ-3: Require confirmation from both buyer and seller before payment is released.
REQ-4: Release payment only once per transaction.
REQ-5: Prevent duplicate payment processing if repeated requests are received.
REQ-6: Display payment failure messages when authorization is denied.
REQ-7: System shall log the payment timestamp for each completed transaction.
REQ-8: System shall record the payment authorization status for each transaction.

4.2 Account Creation

4.2.1 Description and Priority

All users who desire to purchase or sell items will create an account with a login credentials of a username and password. Any other users in “guest mode” will only have the ability to browse listings but nothing else. High Priority.

4.2.2 Stimulus/Response Sequences

- User submits login credentials → System verifies → Access granted or denied
- User updates profile → System validates → Profile gets updated

4.2.3 Functional Requirements

REQ-1: System allows users to register using a unique email address.
REQ-2: System makes sure there's no duplicate email registrations.
REQ-3: Password validation rules.
REQ-4: System allows users to update profile information.
REQ-5: Errors will display for invalid inputs.
REQ-6: Allow authenticated logins and logouts.

4.3 Listing Items

4.3.1 Description and Priority

This feature allows users to create, modify, and manage item listings. This is a high priority feature.

4.3.2 Stimulus/Response Sequences

- Seller inputs how they will get paid if they have not made a sale yet
- Seller photographs items and adds a description, price, etc
- Seller post listing → System validates → Listing status changes to “Available”
- Seller edits listing → System validates → Listing updates
- Seller deleted listing → System verifies status → Listing removed

- Seller selects “create listing” → System displays form → Seller submits details → Listing created

4.3.3 Functional Requirements

REQ-1: System allows sellers to create listings with title, description, category, condition, price, and images.
REQ-2: System creates a unique identifier to each listing (can be a number).
REQ-3: Displays a clear listing status to both buyer and seller.
REQ-4: Make sure to validate required listing fields in req 1
REQ-5: Allows sellers to delete listing not yet sold
REQ-6: Prevent editing of listing marked “Sold”
REQ-7: displays listing status (Available ,Reserved,Sold)

4.4 Buying Items

4.4.1 Description and Priority

This feature is a high priority. This feature allows users to buy the listings that have been put up as for-sale by other users.

4.4.2 Stimulus/Response Sequences

- Buyer inputs how they will make payments if they have not made a purchase yet
- Buyer views item and adds to cart → Buyer purchases item (item will be removed from cart if it gets purchased by another user)
- Buyer and seller view the available areas to meet up at → They agree on one via messaging feature
- They meet up and both users confirm → If both confirm, money moves to seller → If one does not confirm money returns to buyer
- When both confirm transaction completion → System marks transaction “completed” → Payment released

4.4.3 Functional Requirements

REQ1: The system shall allow buyers to purchase listing marked "Available"
REQ2: The system makes sure race condition is in action preventing multiple buyers from purchasing the same listings
REQ-3: System updates listing status to “reserved” upon successful purchase upon.
REQ-4: Prevents reservations if listings status is not labeled as available.
REQ-5: Prevent duplicate purchase requests from repeated clicks.

REQ-6: Log transaction state transitions.

4.5 Map Meetup

4.5.1 Description and Priority

This feature implements a map when a buyer is interested in purchasing to select a spot to meetup. Features will include a request, recommended, and final location. High Priority.

4.5.2 Stimulus/Response Sequences

- Users initiate meetup → System calculates location
- System displays recommended safe public locations
- User selects location → Other party confirms
- System will store this information of location

4.5.3 Functional Requirements

REQ-1: The system shall calculate a midpoint between buyer and seller addresses.
REQ-2: The system shall display suggested public locations near midpoint.
REQ-3: The system shall allow users to request alternate meetup locations.
REQ-4: The system shall store confirmed meetup location for transaction record.

4.6 Feedback

4.6.1 Description and Priority

This feature is to allow for some degree of buyer/seller integrity and is low priority compared to the base features.

4.6.2 Stimulus/Response Sequences

- User completes transaction → User submits rating → System records rating
- User reports listing → System logs report → Admin notified to handle per-case basis

4.6.3 Functional Requirements

REQ-1: The system shall have a 5-star rating system.
REQ-2: The system rating should rate the user not the item itself.
REQ-3: The system should have a review summary.
REQ-4: The review summaries should be allowed to be removed by the reviewee or admin.
REQ-5: The system should be able to report product and seller/buyer post meetings.
REQ-6: The system shall restrict to one rating per transaction.

REQ-7: The system shall display average user rating.
REQ-8: The system shall allow users to report users or listings, an administrator must review said reports.

4.7 Account Deletion

4.7.1 Description and Priority

This feature is to allow for some degree of buyer/seller integrity and is low priority compared to the base features.

4.7.2 Stimulus/Response Sequences

- User selects "Delete Account" → System checks for active pending meetups or active reservations and deletes them
- User confirms deletion → System removes account → User gets logged out and accounts get deleted
- The system sends confirmation notification (email).

4.7.3 Functional Requirements

REQ-1: The system will allow users to delete their account(s).
REQ-2: The system will ask the user a second time if they are sure they want to delete their account.
REQ-3: The system will remove all data from the user after the account is deleted.
REQ-4: The system will provide visual feedback that the account is removed on the UI.
REQ-5: The system will email the user that their account has been deleted.

4.8 Messaging

4.7.1 Description and Priority

This feature is to allow the seller and the buyer to message each other to finalize choosing a meetup location. This is a medium priority feature

4.7.2 Stimulus/Response Sequences

- Seller lists item → Buyer purchase and prompts a chat between the buyer and seller → Buyer and seller choose a location based off of 'Map Meetup' Feature → Transaction is completed (either good or bad) → Chat ends

4.7.3 Functional Requirements

REQ-1: The system will open a chat channel between the two users

REQ-2: The system will maintain the chat channel
REQ-4: The system will send a notification to both users that the chat has opened and will send a message to the recipient every time they receive a message.
REQ-5: The system will maintain chat logs for up to 30 days
REQ-6: No pictures will be shared through messaging

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- Real-time interactions should be instantaneous specifically for confirmed transactions and postings once an item has been sold.
- Ideally the system shall be able to handle a minimum of 500 users at once on the platform, 2,000 listings, 500 transactions.

5.2 Security Requirements

User Authentication:

- Username and Password required to make an account.
- Addresses saved to the database and visible to the personal user but not to anybody else.

Database Security:

- SQL injection
- AES-256 encryption

Password storage:

- Bcrypt
- Argon2
- PBKDF2

5.3 Software Quality Attributes

- Availability: 99% uptime target.
- Reliability: System recovery within 24 hours of critical failure.
- Usability: New users shall complete a purchase within 5 minutes without external. training.
- Maintainability: Code shall follow documented coding standards.
- Scalability: Architecture shall support migration to cloud based infrastructure.

5.4 Business Rules

- Only registered users may create listings or complete purchases.
-
- Payments must be confirmed before meetup coordination.
- Administrators may remove listings suspected of fraud.
- Users with repeated policy violations may be suspended.

6. Other Requirements

- Daily automated database backups shall be performed.
- System logs shall be retained for a minimum of 90 days.
- The system shall comply with applicable data privacy regulations.
- Future mobile application support is outside Version 1.0 scope.

Appendix A: To Be Determined List

- Users with “low” ratings will receive automated warnings and have disclaimers when attempting to buy from them.
- Prioritize the safe places in Harris County with the list over all the other spaces and find a new place if users are not in rage.
- Transactions done in cash.
- Name for the application is subject to change